MIL-W-50219A (AR)
13 February 1978
Superseding
MIL-W-50219 (MU)
30 July 1971

MILITARY SPECIFICATION

WARHEAD SECTION, GUIDED MISSILE, HE, M225, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the assembly and packing of one type of warhead section designated as Warhead Section, Guided Missile, HE, M225.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents.- The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-45782 - Electrical Wiring, Procedure for
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1336

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

S/S UNLESS

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STANDARDS

MILITARY

MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKING DRAWINGS

9215698 - Warhead Section, GM, HE, M225
9226389 - Carton, Packing, Ammunition for Warhead Section (DRAGON)
9226390 - Box, Packing, Ammunition for Warhead Section (DRAGON)

INSPECTION EQUIPMENT DRAWINGS

9246410 - Fixture, test to failure of ring retaining

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. - The components and assemblies shall comply with all requirements specified on drawing 9215698 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Electrical continuity and insulation resistance

3.3.1 Insulation resistance of warhead section, less S&A device. - The resistance between the RED and BLACK wires from the wireway shall be greater than 100 megohms at 100 volts D.C. when tested as specified in 4.5.3.1.

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3.3.2 Continuity of warhead ground circuit.- The resistance between the S&A mounting surface in the boattail of the warhead and the BLACK wire from the wireway shall not exceed 1.0 ohm when tested as specified in 4.5.3.2.

3.3.3 Safety and arming device, GM: M118 (dwg. 9220961).- At time of assembly into the boattail of the warhead section, the S&A Device shall be in the safe, unarmed condition with the warning indicator in the window of the S&A housing showing green when examined as specified in 4.5.4. Before assembly into the warhead section, the S&A Device shall demonstrate its electrical integrity by showing the following resistance measurements when tested as specified in 4.5.4.1, 4.5.4.2 and 4.5.4.3.

3.3.3.1 Open-Circuit Resistance between the INSULATED terminal and the insulated battery pin for the high side of the missile battery. - With the shorting disc (dwg 9318171) REMOVED from the missile battery pins in the S&A Device, the open-circuit electrical resistance between the INSULATED terminal (A) and INSULATED battery pin (C), as indicated in Figure 3, shall read "Open Circuit" when tested using the high scale of an approved "Current-Limited" ohmmeter as specified in 4.5.4.1.

3.3.3.2 Piston actuator continuity. - With shorting disc off, the electrical resistance between the missile battery pins, shown as (C) and (D) on Figure 4, shall measure no less than 1.0 ohm nor more than 2.6 ohms when tested as specified in 4.5.4.2.

3.3.3.3 Insulation resistance between the INSULATED AND UNINSULATED terminals. - With the shorting disc in place on the missile battery pins, the electrical resistance (open-circuit) between the INSULATED (A) and UNINSULATED (B) terminals in the base of the S&A device, as shown in Figure 5, shall read "Open-Circuit" when tested using the high scale of an approved "Current-Limited" ohmmeter as specified in 4.5.4.3.

3.3.4 Capacitance of warhead section.- The capacitance measurement across the black and red wires from the wireway shall be recorded for information only.

3.3.5 Warhead section, M225 (fully assembled)(Dwg 9215698)-

3.3.5.1 Continuity of warhead ground circuit. - The resistance between the base of the S&A device, mounted on the boattail section of the warhead, and the surface of the ogive shall not exceed 1.0 ohm when tested as specified in 4.5.7.1.

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3.3.5.2 Continuity of red wire terminal to insulated S&A.- The resistance between the terminal of the red wire and the insulated terminal on the S&A base shall not exceed 1.0 ohm when tested as specified in 4.5.7.2.

3.3.5.3 Open Circuit Resistance between the red wire terminal and S&A base. - The resistance between the terminal of the red wire and the S&A base shall read "Open Circuit" when tested using the high scale of an approved "Current Limited" ohmmeter as specified in 4.5.7.3.

3.4 Tensile strength of terminal assembly.- Representative samples of wire and terminal lug of terminal assembly (See Dwg. 9215698) shall meet the requirements of MIL-E-45782 para. 3.4.2.1, when subjected to a tensile strength test as specified in 4.5.8 herein.

3.5 Explosive or foreign material.- In the loaded warhead, all exposed surfaces of the liner cone and the interior shroud area, the interior surface of the boattail where the S&A Device seats, and the exterior surface of the warhead body shall be free of foreign material and particles of explosive.

3.6 Weight and center of gravity. - The weight and center of gravity of the fully loaded and assembled warhead section shall be as required by dwg. 9215698.

3.7 Ring, Retaining (dwg. 9276756).

3.7.1 Dimensional stability.- The retaining ring shall not exhibit any deformation when subjected to temperature conditioning of 200°F., plus or minus 5°F., for four (4) hours minimum.

3.7.2 Torque strength. - The retaining ring shall demonstrate an individual torque strength of 300 inch-pounds minimum when tested as specified in 4.5.6."

3.8 First article inspection.- This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship.- The parts and assemblies shall be fabricated, assembled and finished in a thorough, workmanlike manner as defined in a and b.

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a. Fabrication and assembly. - The parts and assemblies shall be complete and shall not have any burrs, sharp edges, dents, gouges or any other undesirable characteristics, signs of damage or poor workmanship.

b. Finishes. - Surface coatings and finishes of parts and assemblies shall be continuous and complete and shall not have scratches, blisters, porosity, corrosion or other signs of defective finish or damage thereto except for superficial scratches on the exterior surface of the warhead body that do not expose the basis metal.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.- The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission.- The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

PART DESCRIPTION	DRAWING	QUANTITY
Warhead Section, GM, HE, M225	9215698	10
Ring, Retaining	9276756	5 from each mold cavity
Screw, Self Locking	9326776	6
Wire Terminal - Ring Tongue Wire size: 26-22	9326777	12
Complete sets of other parts not listed above, pins, rivets, etc.		6 of each
Packing carton	9226389	3
Sealing bag	9226389	3
Packing box	9226390	3

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4.3.2 Inspections to be performed.- See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. - See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50219A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
	Warhead Section GM, HE, M225				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Warhead Section, M225, Less S&A Device (Dwg. 9216775)				
	Examination for defects	10	-	3.2	4.4.2.1
	Insulation resistance - between red and black wires from wireway	10	-	3.3.1	4.5.3.1
	Electrical continuity of warhead/nose crush ground circuit - boattail threaded surface to black wire from wireway	10	-	3.3.2	4.5.3.2
	Tensile strength on terminal lug	6 each	-	3.4	4.5.8
	Capacitance check - between black and red wires from wireway	10	-	3.3.4	4.5.3.3
	Safety & Arming Device, M118 (Dwg. 9220961)				
*	S&A device safe	10	-	3.3.3	4.5.4
	Presence of lead charge	10	-	3.3.3	4.5.4
	Resistance measurement on S&A device: Open - circuit resistance between insulated terminal and insulated missile battery pin (with shorting disc removed)	10	-	3.3.3.1	4.5.4.1

NOTES:

* Critical Defect

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50219A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
	Continuation Sheet				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Piston actuator continuity	10	-	3.3.3.2	4.5.4.2
	Insulation resistance between insulated terminal and uninsulated terminal (with shorting disc in place)	10	-	3.3.3.3	4.5.4.3
	Warhead Section, M225 (Fully Assembled) Dwg. 9215698				
	Examination for defects	10	-	3.2	4.4.2.3
	Weight	10	-	3.6	4.5.5.1
	Longitudinal center of gravity	10	-	3.6	4.5.5.2
	Continuity of warhead ground circuit	10	-	3.3.5.1	4.5.7.1
	Continuity of red wire terminal to insulated S&A terminal	10	-	3.3.5.2	4.5.7.2
	Open circuit resistance of red wire terminal to S&A base	10	-	3.3.5.3	4.5.7.3
	Retaining Ring (Dwg. 9276756)				
	Examination for defects	*	-	3.7.1	4.4.2.4
	Torque strength test	*	-	3.7.2	4.5.6/Dwg 9246410

NOTES:

* 5 from each mold cavity

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50219A (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Continuation Sheet				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Screw, Self-Locking (Dwg. 9326776)				
	Examination for defects	6	-	3.2	4.4.2.5
	Wire Terminal-Ring Tongue Wire Size 26-22 (Dwg. 9326777)				
	Examination for defects	12	-	3.2	4.4.2.6
	Other Components				
	Examination for defects	6 each	-	3.2	4.4.2.7
	Carton, Packing Prior to Sealing Bag (Dwg. 9226389)				
	Examination for defects	3	-	3.2	4.4.2.8
	Bag, Prior to Insertion into Packing Box (Dwg. 9226389)				
	Examination for defects	3	-	3.2	4.4.2.9
	Packing Box, Prior to Sealing (Dwg. 9226390)				
	Examination for defects	3	-	3.2	4.4.2.10
	Box, Packing, Ammo, for Warhead Section GM, HE, M225, Sealed, (Dwg. 9226390)				
	Examination for defects	3	-	3.2	4.4.2.11

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation.- Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of loaded warheads shall contain:

a. Loaded warheads, M224, from not more than one interfix number from one manufacturer.

b. Nose crush switch assemblies from not more than one interfix number from one manufacturer.

c. Retaining rings from not more than one interfix number from one manufacturer.

d. S&A devices from not more than one interfix number from one manufacturer.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50219A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9216775
4.4.2.1	Warhead Section, GM, HE, M225 Less S&A Device				NEXT HIGHER ASSEMBLY 9215698
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Examination for presence of explosive or foreign material outside the explosive cavity		100%	3.5	Visual
102	Marking		100%	3.2	Visual
103	Proper assembly - parts missing, loose or improperly assembled		100%	3.2	Visual
104	Insulation resistance - between red and black wires from wireway		100%	3.3.1	4.5.3.1
105	Electrical continuity of warhead/nose crush ground circuit - boattail, threaded surface to black wire from wireway		100%	3.3.2	4.5.3.2
106	Tensile strength on terminal lug	*Each	4.0%	3.4	4.5.8
107	Capacitance check - between black and red wires from wireway		100%	3.3.4	4.5.3.3
Minor					
201	Gap and step at ogive to warhead body joint		0.65%	3.2	Gage
202	Gap between wireway and body		0.65%	3.2	Gage
203	Protective finish and touched-up surfaces		0.65%	3.9.b	Visual
204	Evidence of poor workmanship		1.00%	3.9	Visual

NOTES:

* In accordance with MIL-STD-105, Inspection Level 1.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50219A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9220961
4.4.2.2	Safety & Arming Device, M118				NEXT HIGHER ASSEMBLY 9215698
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	S&A Device safe		100%	3.3.3	4.5.4
Major					
101	Presence of lead charge		100%	3.3.3	4.5.4
102	Resistance measurement on S&A device: Open circuit resistance between insulated terminal and insulated missile battery pin (with shorting disc removed)		100%	3.3.3.1	4.5.4.1
103	Piston actuator continuity		100%	3.3.3.2	4.5.4.2
104	Insulation resistance between insulated terminal and uninsulated terminal (with shorting disc in place)		100%	3.3.3.3	4.5.4.3

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50219A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Warhead Section, GM, HE, M225 (Fully Assembled)				9215698
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Torque on retaining ring		0.40%	3.2	Gage
102	S&A screw torqued		0.40%	3.2	Gage
103	Pin missing		0.40%	3.2	Visual
104	Turn part 360°, end-over-end to detect any loose parts		100%	3.2	Visual/Manual
105	Marking - part no., lot no., etc.		100%	3.2	Visual
106	Weight		100%	3.6	4.5.5.1
107	Longitudnal center of gravity		100%	3.6	4.5.5.2
108	Protective finish improper or inadequate		0.40%	3.2	Visual
109	Continuity of warhead ground circuit		100%	3.3.5.1	4.5.7.1
110	Continuity of red wire terminal to insulated S&A terminal		100%	3.3.5.2	4.5.7.2
111	Open circuit resistance of red wire terminal to S&A base		100%	3.3.5.3	4.5.7.3
Minor					
201	Outer nose tip surface of ogive not free of paint		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.9	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50219A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Retaining Ring				9276756
					NEXT HIGHER ASSEMBLY 9215698
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter		0.40%	3.7.1	Gage
102	Minor diameter		0.40%	3.7.1	Gage
103	Flange inside diameter		0.40%	3.7.1	Gage
104	Torque strength test		*	3.7.2	4.5.6 /Dwg. 9246410
Minor					
201	Evidence of poor workmanship		1.00%	3.9	Visual

NOTES: * A sample of ten (10) retaining rings shall be selected from each day's continuous production run from each mold cavity.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50219A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Screw, Self-Locking		9326776
			NEXT HIGHER ASSEMBLY 9215698

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Locking element loose or missing		0.40%	3.2	Visual
102	Locking element area does not pass through ring gage as per note 4 of Dwg. 9326776		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.9	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50219A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Wire Terminal - Ring Tongue Wire Size: 26-22		9326777
			NEXT HIGHER ASSEMBLY 9215698

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of hole		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
Minor					
201	Total length, max.		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.9	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Other components		See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spec or Dwg. No. / Document Nomenclature				
	MIL-R-2423/6 - C402H — Rivet				
	MS16562-2 — Pin				
	The contractor or vendor shall supply a certificate that the materials used, conform to the drawing requirements				
Critical	None defined				
Major					
101	Dimensions improper		0.40%	3.2	Gage
Minor					
201	Protective finish inadequate (if applicable)		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.9	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Carton, Packing, Prior to Sealing Bag		9226389
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Protector missing		0.40%	3.2	Visual
102	Spacer missing		0.40%	3.2	Visual
103	Sleeve missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.9	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Bag, prior to Insertion into Packing Box		9226389
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Bag damaged (cuts, tears, holes)		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		1.00%	3.9	Visual

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DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 71 which is Obsolete.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50219A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9226390
4.4.2.10	Packing Box, Prior to Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Contents improperly packed		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		1.00%	3.9	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50219A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9226390
4.4.2.11	Box, Packing, Ammo, for Warhead Section, GM, HE, M225, Sealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	DOD symbol missing, incorrect or illegible		0.40%	3.2	Visual
102	Contents loose when tilted		0.40%	3.2	Visual
103	Strapping missing, broken or loose		0.40%	3.2	Visual
Minor					
201	Board broken or split		0.65%	3.2	Visual
202	Marking missing, misleading or illegible		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.9	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Testing.- Testing is described in the First Article and Quality Conformance tables.

4.4.4 Inspection equipment.- The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes.- Compliance with all requirements of section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, sub-assemblies and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual, mechanical and electrical inspections.- The visual manual, mechanical and electrical inspections and tests shall verify compliance with the requirements of sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Molded components.- At the start of production from a new mold, or repair to a mold or cavity, consecutively produced units from each mold or cavity shall be physically and dimensionally examined until four (4) consecutive units meet all applicable drawing requirements. After adjustment or repair, those dimensions affected by the adjustment or repair shall be examined. During production, the molding procedures shall be monitored for uniformity of molding practices and quality conformance of product. When a Classification of Defects under 4.4.2 is specified for a molded component, lot acceptance shall be in accordance thereto and will also serve as a continuing check on the molding process. Suspected deviations in the mold or product shall be examined to determine if corrective action or mold repair is required.

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4.5.3 Electrical check-outs on warhead section, less S&A Device (dwg. 9216775). - Electrical resistance and continuity measurements shall be made on the warhead section with nose crush switch assembled to the loaded warhead, but prior to assembly of the safety and arming device in the boattail of the warhead section.

4.5.3.1 Insulation resistance.- The electrical resistance shall be measured between the Red wire and the Black wire from the wireway, as shown in Figure 1, using a megohmmeter as specified in accordance with the provisions of 4.4.4. The test shall be conducted utilizing a minimum of 100 volts D.C. Any warhead section which fails the minimum resistance requirement specified in 3.3.1 shall be classed defective.

4.5.3.2 Continuity. - This electrical resistance test for continuity shall be measured between the S&A Device seating surface on the boattail of the warhead and the Black wire from the wireway, as shown in Figure 2. An approved milliohmmeter, as specified in accordance with the provisions of 4.4.4, may be used. Any warhead section which exhibits a resistance value not in accordance with the requirement specified in 3.3.2, shall be classed defective.

4.5.3.3 Capacitance. - The capacitance measurement shall be taken across the black and red wires from the wireway. The test current and voltage shall be limited to 40 microamperes, max., and 100 volts direct current (v.d.c.) max. Test equipment shall be as specified in 4.4.4.

4.5.4 Unarmed Condition of the Safety and Arming Device, GM: M118 (dwg. 9220961). - Prior to assembly in the boattail of the warhead section, the safety and arming device shall be visually examined to assure a safe handling condition. The warning indicator shall be viewed through the window in the housing of the S&A Device and a safe condition confirmed by observing a green indicator. If the indicator is red, partially red, or total green indicator cannot be observed, the S&A shall be classed defective and shall be considered armed and hazardous. Under these circumstances, the defective S&A shall be set aside for future disposition (see 4.5.4.4 below). At the same time, the presence of the lead charge in the cup at the top of the S&A housing shall be verified visually. If not present, the S&A Device shall not be used in assembly of the warhead section. Unarmed S&A Devices shall then be checked out electrically as specified in 4.5.4.1, 4.5.4.2 and 4.5.4.3 to the requirements of 3.3.3. These measurements can be facilitated by using mating plug connectors with lead wires to the ohmmeters, which can plug into the respective receptacles and pins in the housing base of the S&A Device.

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4.5.4.1 Open-Circuit resistance measurement (See Figure 3)- Between the insulated terminal and the insulated Battery Pin for the High Side of the Missile Battery (both located in the base of the S&A Device) (See 3.3.3.1). - Prior to making this electrical measurement, remove the shorting disc. To measure open circuit resistance connect the leads of the current-limited ohmmeter across the insulated terminal, shown as (A) on Figure 3, and the insulated battery pin, shown as (C). The ohmmeter used shall have a higher scale representing at least 1000 ohms full scale and shall limit the current to a maximum of one (1) milliampere as specified in accordance with the provisions of 4.4.4. The resistance reading shall indicate an "open circuit on the high scale" as specified in requirement 3.3.3.1.

WARNING: The electrical resistance measurements may be dangerous and should be performed behind a suitable barricade.

4.5.4.2 Piston Actuator Resistance. - With shorting disc off, measure the continuity of the piston actuator circuit by connecting the leads of the current-limited milliohmmeter across the insulated and uninsulated missile battery pins shown as (C) and (D) respectively on Figure 4. The continuity resistance exhibited in this test shall be within the limits specified in requirement 3.3.3.2.

WARNING: The electical resistance measurements may be dangerous and should be performed behind a suitable barricade.

4.5.4.3 Insulation Resistance between the Insulated and Uninsulated terminals in the Base of the S&A Device. - Before making this electrical check, replace the shorting disc (9318171) on the missile battery pins. To measure the open-circuit resistance, connect the leads of the current-limited ohmmeter across the Insulated and Uninsulated terminals in the base of the S&A device. These connections are shown as (A) and (B) respectively on Figure 5. The resistance reading shall indicate an "open circuit on the high scale" as specified in requirement 3.3.3.3.

WARNING: The electrical resistance measurements may be dangerous and should be performed behind a suitable barricade.

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ELECTRICAL CHECK-OUTS ON WARHEAD SECTION, LESS S&A DEVICE

(After assembly of Nose Crush Switch to Warhead, M224)
(paragraph 4.5.3)

Insulation Resistance (4.5.3.1)

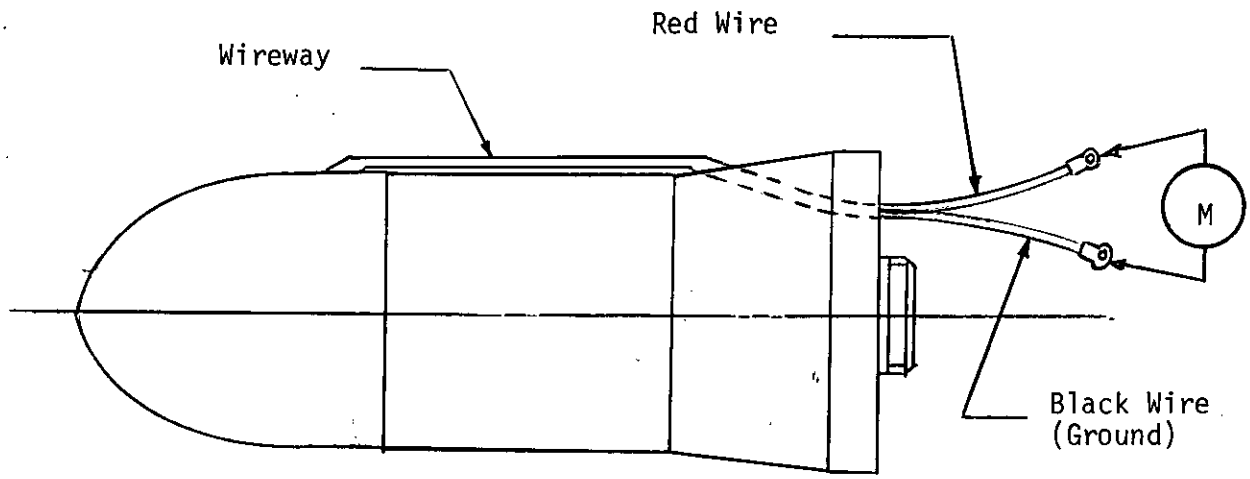


FIGURE 1

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ELECTRICAL CHECK-OUTS ON WARHEAD SECTION, LESS S&A DEVICE

(After assembly of Nose Crush Switch to Warhead, M224) (paragraph 4.5.3.)

Continuity (4.5.3.2)

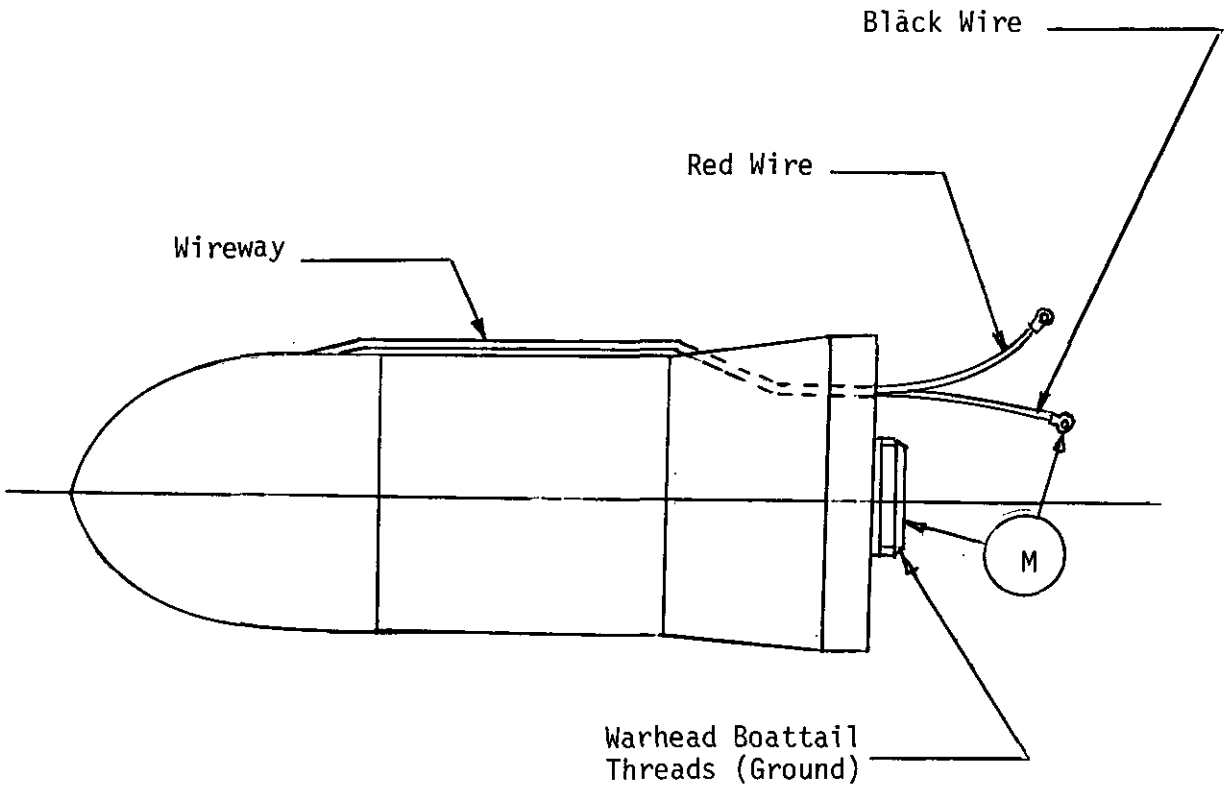


FIGURE 2

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ELECTRICAL CHECK-OUTS ON S&A DEVICE, M118
(Before Assembly to Warhead Section M225)

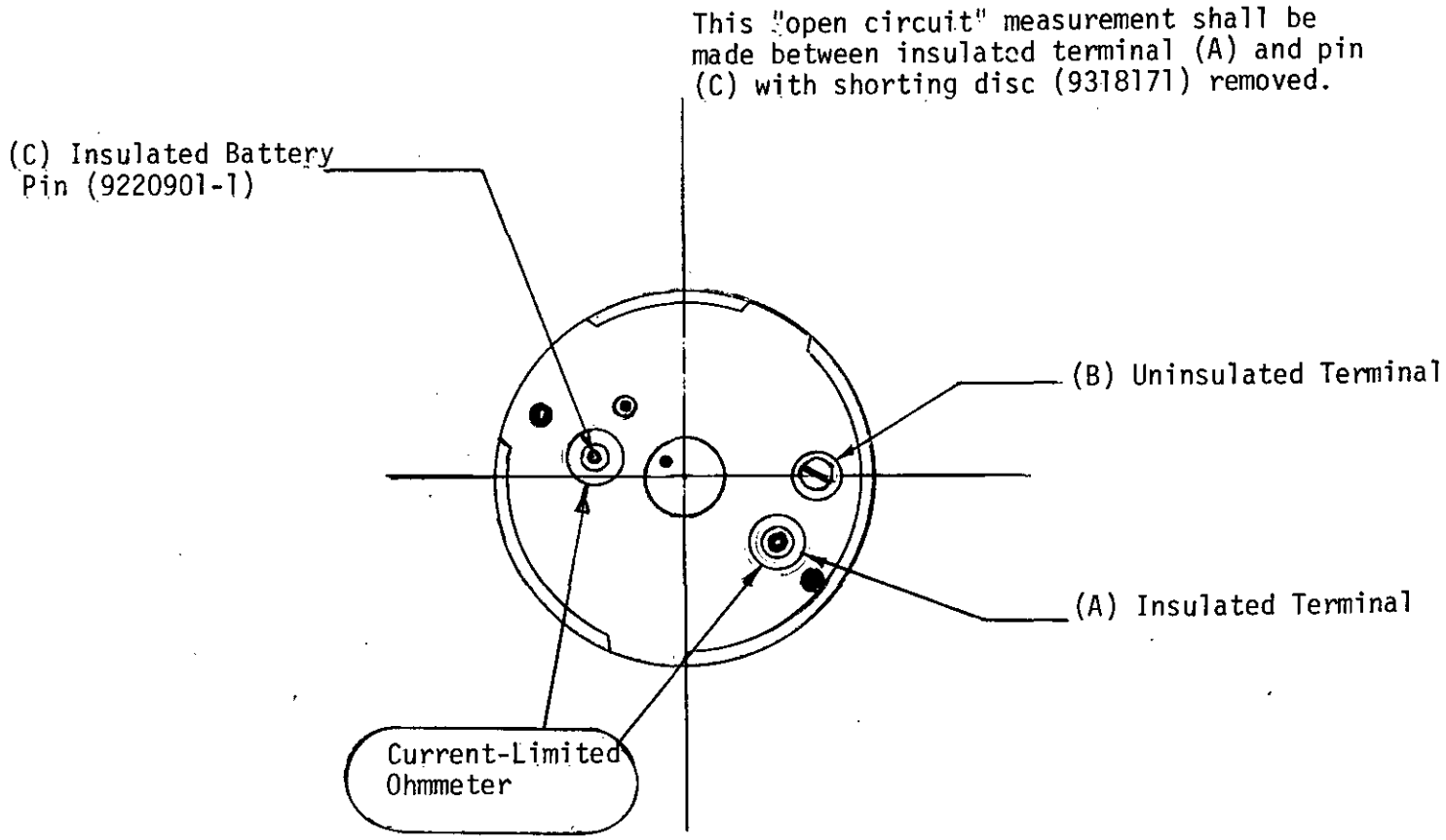


FIGURE 3 (3.3.3.1 and 4.5.4.1)

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ELECTRICAL CHECKOUTS ON S&A DEVICE, M118
(Before Assembly to Warhead Section M225)

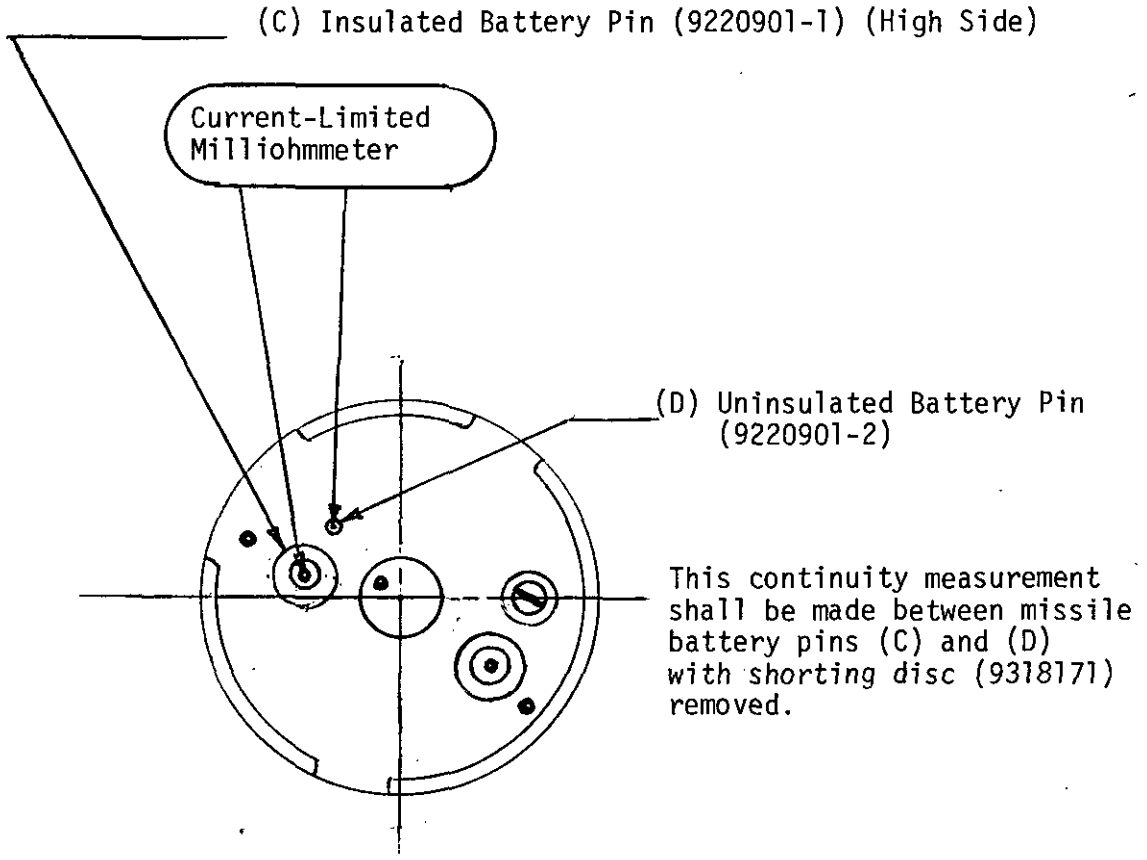


FIGURE 4 (3.3.3.2 and 4.5.4.2)

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ELECTRICAL CHECK-OUTS ON S&A DEVICE, M118
(Before Assembly to Warhead Section M225)

With shorting disc on this "open circuit" measurement shall be made between insulated terminal (A) and uninsulated terminal (B)

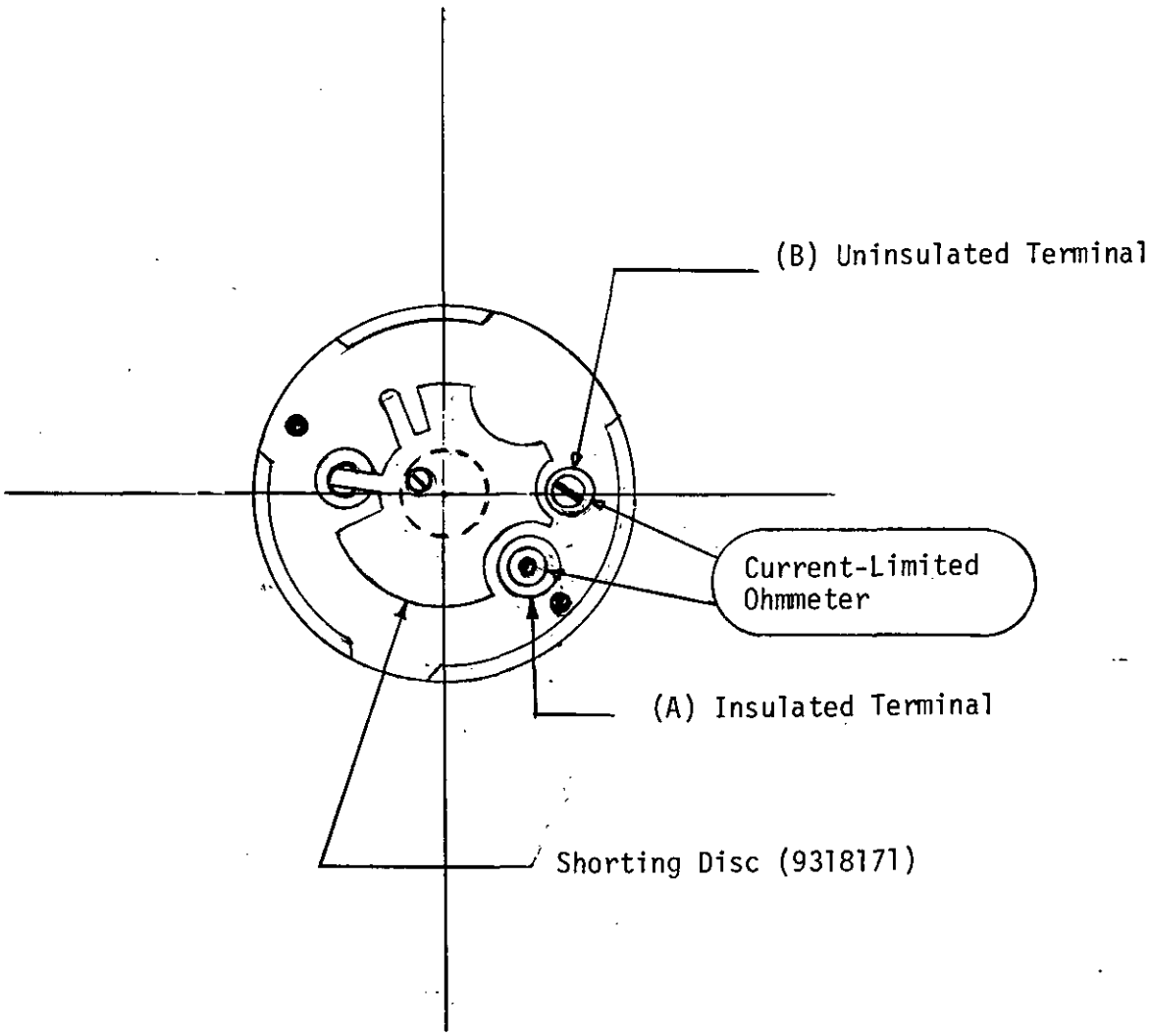


FIGURE 5 (3.3.3.3 and 4.5.4.3)

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4.5.4.4 Disposition of Defective S&A Devices (dwg. 9220961). Any S&A Device which is indicated to be armed or fails in any way to demonstrate the required electrical characteristics and resistance measurements as specified in 3.3.3, 3.3.3.1, 3.3.3.2, and 3.3.3.3 when tested as prescribed in 4.5.4, 4.5.4.1, 4.5.4.2 and 4.5.4.3 shall NOT be assembled into a warhead section, but shall be set aside for future disposition in accordance with prescribed Safety Standard Operating Procedures.

4.5.5 Weight and Center of Gravity of the Warhead Section, GM, HE, M225 (dwg. 9215698).

4.5.5.1 Weight of Assembly.- The fully loaded and assembled warhead section shall be weighed on a balance scale accurate to .005 lb. Any assembly which does not satisfy the weight requirement for the warhead section, as specified in 3.6, shall be classed defective.

4.5.5.2 Center of Gravity. - The warhead section shall be mounted in the center of gravity fixture. The location of the center of gravity shall be measured in the longitudinal axis. Any assembly which does not comply with the requirement, as specified in 3.6, shall be classed defective.

4.5.6 Torque strength of ring, retaining.- The ring, (dwg. 9276756) shall be assembled to a post, dwg. 9246410. A torque adapter, dwg. 9246410, assembled to a torque wrench shall be positioned over the ring, and shall be torqued to the specified requirement of 3.7.2. Failure of one or more rings to meet the minimum strength requirement shall reject the day's production represented by the sample.

4.5.7 Electrical Check-outs on Warhead Section, With S&A Device Installed (Dwg 9215698)

4.5.7.1 Continuity of Warhead Ground Circuit.- Before making this electrical check, replace the shorting disc (9318171) on the missile battery pins. The continuity shall be measured between the base of the S&A device and the surface of the ogive. An approved ohmmeter (current-limited), as specified in accordance with the provisions of 4.4.4 shall be used. Any warhead assembly which exhibits a resistance value not in accordance with the requirement specified in 3.3.5.1., shall be classed defective.

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4.5.7.2 Continuity of Red Wire Terminal to Insulated S&A Terminal. - Before making this electrical check, replace the shorting disc (9318171) on the missile battery pins. To measure this closed circuit resistance, attach one lead to the terminal of the red wire and the other lead to the insulated terminal on the base of the S&A. An approved ohmmeter (current-limited) as specified in accordance with the provisions of 4.4.4 shall be used. Any warhead assembly which exhibits a resistance value not in accordance with the requirement specified in 3.3.5.2 shall be classed defective.

4.5.7.3 Open Circuit Resistance between the Red Wire Terminal and S&A Base. - Before making this electrical check, replace the shorting disc (9318171) on the missile battery pins. To measure this open circuit resistance, attach one (1) lead to the terminal of the red wire and the other lead to the base (access cover) of the S&A. An approved ohmmeter (current limited) as specified in accordance with the provisions of 4.4.4 shall be used. Any warhead assembly which exhibits a resistance value not in accordance with the requirement specified in 3.3.5.3 shall be classed defective.

WARNING: The above electrical measurements in 4.5.7.1, 4.5.7.2, and 4.5.7.3 may be dangerous and should be performed behind a suitable barricade.

4.5.8 Tensile strength of terminal assembly.- The representative samples of the terminal lugs and electrical wire which has the end of wire tin dipped and the terminal lugs crimped shall be subjected to a tensile strength test in accordance with MIL-E-45782, para. 3.4.2.1 and 4.6.2.1.2 except the speed ot the holding head shall not exceed 10 inches per minute (Destructive test).

5. PACKAGING

5.1 Preservation and Packaging

5.1.1 Level C. - The warhead section shall be preserved and packaged in accordance with dwg. 9226389.

5.2 Packing

5.2.1 Level C. - The warhead section shall be packed in accordance with dwg. 9226390.

5.3 Marking. - Marking of the wood packing box shall be in accordance with dwg. 9226390. The carton and bag shall be marked in accordance with dwg. 9226389.

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6. NOTES

6.1 Intended use. - The Warhead Section, GM, HE, M225, is a high explosive, shaped-charge warhead with fuzing system - Fuze: M815 - intended for assembly to the Dragon Missile System, M47.

6.2 Ordering Data.- See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals.- See MIL-A-48078. Submit equipment designs, as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Submission of test data. - When required by incorporation of DD Form 1423 into the contract, all test data required in Section 4 of this specification shall be submitted in accordance with Data Item Description DI-R-1724. In addition to the normal distribution of records, one copy of all test data shall be forwarded to each of the following:

Commander
US Army Armament Research and Development Command
ATTN: DRDAR-QAR-M and
DRDAR-LCU-D
Dover, New Jersey 07801

6.5 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.6 Definitions

6.6.1 Contractor. - Unless otherwise specified, the word "Contractor", as used throughout this specification, shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government-Owned, Contractor-Operated (GOCO) and Government-Owned, Government-Operated (GOCO) plants.

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6.7 Changes from previous issue. - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Preparing Activity:
Army-AR

Project Number: 1336-A295

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MIL-W-50219A WHD, GM, HE, M225, ASSEMBLING & PACKING

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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